17 Bridging the Gender Digital Divide Using Analog and Digital Technologies in Rural Kenya

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17.1 Introduction

Information and communication technologies (ICT) can help reduce gender disparities—particularly the digital divide—supporting progress toward Sustainable Development Goal 5 on gender equality. Ensuring women's equitable access to technology and societal participation requires coordinated action from governments, industry, and civil society. Technology holds promise for empowering women in rural areas by improving access to agricultural knowledge. However, gendered disparities persist across access, socioeconomic status, and remoteness. ICTs can help narrow these gaps and support progress toward sustainable development goals [1]. This chapter highlights an agricultural change agent and radio presenter who enhances women's access to farming knowledge using radio, mobile phones, and video animations. Their work fosters lifelong learning and amplifies women's voices through reciprocal feedback loops. Women are encouraged to interact with content and share experiences via mobile and radio call-ins. This feedback-driven learning model turns the change agent into a learner and the audience into educators. Their adaptability—choosing optimal broadcast times, responding to suggestions, and engaging in field interactions—is key to empowering women. The agent's evolving methods are just as critical as the women's own learning in enhancing information access.

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17.1.1 Lifelong Learning

Lifelong learning (LLL) spans formal, informal, and non-formal modes, fostering ongoing, self-directed knowledge across life. Closely linked to adult education, LLL thrives through change agents who share information and maintain engagement—such as via radio call-in programs. In this context, women not only learn but also educate the change agent by sharing lived experiences. This reciprocal interaction forms a feedback loop, enriching face-to-face learning while harnessing ICTs like mobile phones and analog radio. The university-based initiative Scientific Animations Without Borders (SAWBO) illustrates how ICTs—especially mobile-delivered educational videos—can dramatically expand access to knowledge while reducing the costs of scaling their reach [2]. SAWBO produces free, multilingual video animations for marginalized populations, available across digital and analog platforms. Its video library, covering agriculture, health, economics, peace and justice, women's empowerment, and climate change, is accessible via USB drives, social media (WhatsApp, Facebook), YouTube, supporting websites, TV and radio broadcasts, training workshops, and an Android-based app [3].

Scientific Animations Without Borders (SAWBO) applies a systems-based approach to develop and scale educational video content, promoting sustainable livelihood outcomes [4]. Its library hosts 200+ multilingual animations on 150+ topics in over 300 language variants, reaching 70+ million viewers across 130 countries [5]. By ensuring broad accessibility, SAWBO fosters lifelong learning and practical solutions to locally recognized challenges. This chapter examines how blending analog and digital technologies—especially mobile phones and radio—can bridge the gender digital divide in rural Kenya. It highlights barriers to women's digital access and proposes mobile-based strategies that advance gender equity, in line with SDG 5.

17.1.2 Goal 5: Achieve Gender Equality and Empower All Women and Girls

ICTs are key drivers in achieving the 17 UN Sustainable Development Goals (SDGs), adopted in 2015 to protect the planet and ensure equitable prosperity by 2030 [6]. Gender equity is central to this agenda, with SDG 5 and its 54 indicators targeting the elimination of discrimination and violence against women while promoting their empowerment. Sustainable Development Goal (SDG) 5 emphasizes gender equality in digital access. As Harvey et al. [7] highlight, personal mobile phone ownership enhances women's empowerment and productivity. It provides autonomy across personal and professional spheres, making mobile ownership a key metric under SDG indicator 5.b.1—proportion of mobile phone owners by sex. Likewise, indicator 5.c.1 tracks national systems that publicly allocate resources for gender equality and women's empowerment (Table 17.1).

Table 17.1 Goal 5 Gender digital divide targets and indicators ⏎

Goal 5 Gender Equality	Targets	Indicators
Goal 5: Achieve gender equality and empower all women and girl	5.B Enhance the use of enabling technology, in particular information and communications technology, to promote the empowerment of women	Indicator 5.b.1. is the Proportion of individuals who own a mobile telephone, by sex
	5.C Adopt and strengthen sound policies and enforceable legislation for the promotion of gender equality and the empowerment of all women and girls at all levels	Indicator 5.c.1 is the Proportion of countries with systems to track and make public allocations for gender equality and women's empowerment.

Adapted from United Nations [6]

17.1.3 Background on the Gender Digital Divide

Technological revolutions have redefined societal structures and value chains. As Hamburg and Lütgen [8] note, the digital divide reflects disparities in access and engagement with ICTs—not just in usage, but in skills and tools. This exclusion limits informed decision-making and obstructs competitive advancement [1].

The digital divide reflects global socioeconomic disparities [9, 10], with access and connectivity as key concerns. Beyond these, the divide spans four dimensions: limited online access due to demographics (information divide); gaps in digital proficiency (skills divide); unequal access to education and jobs (economic divide); and exclusion of certain groups from digital engagement (demographic divide) [11, 12]. The digital divide refers to inequalities that separate societies and nations into those with—and without—digital access, skills, and knowledge, leading to uneven use of technologies and benefits [13]. Initially seen as a matter of access to ICTs, it now includes disparities in digital engagement, usage, and outcomes due to broader social inequalities [13].The digital divide is commonly structured in three levels [13]:

Access Divide: Unequal access to devices and internet, shaped by factors like gender, age, ethnicity, and global privilege.

Skills Divide: Differences in digital proficiency required for meaningful use.

Outcomes Divide: Gaps in offline benefits from ICT use, including civic, educational, and social participation. Smartphone growth has also introduced newer divides and knowledge disparities [13].

Mutula categorizes the digital divide into four key dimensions [14]:

1. **Social divide:** Cultural disparities in digital access and usage across states.
2. **Economic divide:** Barriers from poverty and financial limitations that constrain digital benefits.
3. **Linguistic divide:** Gaps due to unfamiliarity with the language of digital tools and content.
4. **Content divide:** Mismatch between digital content and the needs of excluded populations, often stemming from non-inclusive development processes.

17.2 The Gender Digital Divide in Rural Kenya and Communities of Practice

The gender divide in rural Kenya stems from socioeconomic barriers, cultural norms, and infrastructural gaps [15]. Traditional roles and societal expectations often limit women's engagement with technology, framing digital literacy as male-dominated [16]. Poor internet connectivity, unreliable electricity, and costly devices further restrict access, particularly for rural women [10]. Bridging this divide requires inclusive policies—such as gender-sensitive education, affordable digital access, and shifts in cultural perceptions—to promote women's digital empowerment. Mobile services offer a unique lens into ICTs and gender disparities—particularly in agriculture, smartphone ownership, and education [17]. In Kenya, men often own smartphones and make agricultural decisions as household heads [17]. Santosham and Lindsey [18] highlight affordability as a key barrier, with women less likely to own smartphones. Additional constraints for rural women include low digital literacy, poor device quality, negative internet perceptions, limited learning time, and unstable income sources [10, 19, 20].

17.2.1 The Role of Analog Technologies in Digital Inclusion

Diverse analog and digital channels—such as extension services, field days, print media, TV, radio, and the internet—play a vital role in enhancing agricultural productivity in Kenya [21]. Radio in local languages is especially effective for reaching subsistence farmers with limited English fluency or print literacy [21]. Integrating analog tools with digital learning is essential for agricultural education in rural Kenya, where internet access is often limited. Resources like radio, print media, and community forums play a crucial role in bridging the digital divide by delivering accessible agricultural knowledge to local farmers. Analog tools remain pivotal in agricultural education, especially in rural Kenya [22]. Radio stations like *Shamba Shape Up* and *Farmers' Forum* provide real-time guidance on weather, farming, pest control, and

markets [21]. With widespread accessibility via phones and traditional radios, these broadcasts reach remote areas lacking internet connectivity. Printed materials remain essential analog tools for agricultural education, offering step-by-step guidance in local languages to support self-paced learning and complement digital media. However, low print-literacy—particularly among women—significantly constrains their impact [23]. Farmers Field Schools and community meetings provide crucial platforms for peer-to-peer, hands-on agricultural learning [24], enabling real-time exchange of experiences and techniques. These forums foster social learning and a community of practice that supports inclusive knowledge transfer and agricultural advancement [25]. Blending analog tools—radio, printed guides, and community events—with digital methods bridges the rural digital divide by offering accessible, low-cost learning alternatives. Initiatives like SAWBO videos and radio programming illustrate how such integration enhances agricultural education, especially for women. Yet, literacy and financial hurdles hinder adoption. Gender-sensitive policies and community-driven strategies are vital to ensure equitable digital access and empower women in lifelong learning.

17.2.2 Hotspotting and Mobile Technologies as a Tool for Bridging the Divide

In Kenya, 49.5% of mobile owners are women and 50.5% are men, with ownership rates at 43.2% for women and 45.5% for men overall. Urban areas show higher mobile ownership (59%) than rural regions (36%). Notably, women in low- and middle-income countries like Kenya are 13% less likely to own smartphones, highlighting persistent gender gaps in digital access [7].

In Kenya, mobile phone hotspotting offers a cost-effective way to expand ICT access, especially for women in rural areas. By sharing mobile data through personal or community networks, hotspotting helps close the internet access gap for underserved populations. In rural Kenya, mobile hotspotting —using phones as Wi-Fi routers—offers an alternative to fixed broadband, leveraging widespread cellular networks to extend internet access. Despite challenges like cost, data limits, signal security, and gender gaps [26], solar and alternative recharging methods can enable connectivity even in unelectrified regions [27, 28].

Despite its potential, mobile hotspotting in Kenya faces hurdles: costly data limits usage among low-income households, patchy rural network coverage reduces effectiveness, and women's limited smartphone access—often relying on others for connectivity—excludes them from full digital participation [26]. Addressing hotspotting barriers is key to narrowing Kenya's digital divide, especially in rural and marginalized communities. Shared mobile access supports inclusion, education, and women's empowerment. Scaling such initiatives alongside affordable internet policies can fast-track progress on gender and digital equity. To maximize hotspotting's impact on digital

inclusion, stakeholders can offer subsidized data plans, invest in digital literacy, and promote gender-inclusive mobile ownership policies. Supporting locally relevant content and raising awareness—such as pooling resources for a shared hotspot-enabled phone—can further extend access in marginalized communities (Table 17.2).

Table 17.2 Reducing the digital divide through hotspotting ⏎

Reducing the Gender Digital divide Using Hotspotting	Examples Indicating Gendered Access to Hotspotting
Expanding internet access for marginalized groups	Many women, small-scale farmers, and entrepreneurs in rural Kenya struggle with internet access due to the high cost of personal data plans and the absence of home Wi-Fi connections. Phone hotspotting allows individuals to share data with family members, friends, or community groups, making digital access more affordable and widely available.
Enhancing digital learning and capacity building and lifelong learning	Hotspotting plays a crucial role in educational initiatives, including digital literacy programs targeted at women and youth. Community groups, schools, and training centers often rely on a single mobile device to provide internet access to multiple learners, enabling them to access online courses, research materials, and digital skill-building content.
Empowering women and rural entrepreneurs	Women in rural areas face challenges such as ownership of devices and internet due to financial constraints, cultural norms and lack of internet infrastructure. With hotspotting, a single mobile device can provide multiple women with access to essential services, such as mobile banking, e-commerce platforms, and agricultural market information. This can lead to economic empowerment.
Supporting agricultural extension and information dissemination	Farmers in Kenya increasingly rely on mobile technology for agricultural extension services, weather forecasts, and market price updates. Phone hotspotting allows farmer groups to collectively access educational content. Through phone hotspotting farmers experience knowledge sharing even when they do not own internet connection.
Facilitating telehealth and mobile health services	Apart from agriculture mobile hotspotting is also used in the health sector. It enables access to telehealth services in rural areas where health infrastructure is limited thus improving health outcomes. Women and families can access virtual consultations, maternal health advice, and public health updates through shared internet access.

17.2.3 Communities of Practice a Framework for Supporting Women's Digital Empowerment

Communities of Practice (CoPs) are key to women's digital empowerment, enabling collaborative learning, knowledge exchange, and skills development in supportive settings [29]. Through shared goals, CoPs foster peer learning, mentorship, and localized solutions that help women overcome barriers to digital access and use [30, 31]. Scientific Animations Without Borders (SAWBO) uses multilingual, culturally tailored videos to empower women in agriculture, health, and entrepreneurship, bridging digital divides—even in low-literacy settings [32]. Through WhatsApp and community screenings, SAWBO fosters Communities of Practice that promote peer learning and digital skill-building, enabling women to become changemakers and sustain empowerment efforts [33].

17.3 Case Study: Using Analog and Digital Tools to Reduce Gender Digital Divide in Kenya

Agriculture is central to livelihoods in Kenya, yet rural farmers—especially women—struggle to access vital knowledge. Juma, a radio presenter and agricultural change agent, uses radio and SAWBO videos to promote lifelong learning and practical farming skills. His approach fosters continuous learning and highlights the role of analog and visual tools in closing information gaps. Juma plays a vital role in educating women farmers who lack access to formal training or digital tools. His radio show reaches offline communities, providing expert insights and peer learning. By incorporating audience feedback, Juma fosters a two-way learning process, making his approach responsive and impactful in rural agriculture. To suit farmers' daily routines, Juma airs his show at 6:00 p.m.—a time identified through community feedback as most accessible for women listeners. This trial-and-error approach reflects lifelong learning, as Juma continually adapts to audience needs.

Juma's radio show shares vital insights on sustainable farming, pest control, crop rotation, and weather updates. To enhance understanding, he supplements broadcasts with SAWBO video animations—engaging visuals that simplify complex techniques and show farmers how to apply them on the ground. By combining radio and SAWBO animations, Juma addresses diverse learning needs—radio delivers core content, while visuals reinforce key techniques. For instance, he explains crop planting on-air, then follows up with step-by-step animations, boosting comprehension and practical adoption. Juma fosters lifelong learning by encouraging farmers—especially women—to engage actively with new knowledge. They share experiences, reflect on techniques, and adapt them in practice, creating a dynamic feedback loop that reinforces continuous learning.

Despite the wealth of digital resources, rural farmers in Kenya face barriers like poor connectivity, high data costs, and limited devices. Radio remains a vital analog tool for reaching those offline, while SAWBO animations offer engaging visuals that reinforce agricultural training. Radio's low cost and wide reach make it ideal for resource-limited rural areas. Juma's broadcasts, combined with SAWBO animations, extend visual learning to those without digital access—bridging infrastructure gaps and ensuring inclusive agricultural education. Juma exemplifies lifelong learning by adapting his methods to listener feedback—from adjusting his airtime to refining the use of SAWBO animations. His role extends beyond sharing content; he models continuous learning through practice.

Juma's journey as a change agent reflects lifelong learning, marked by ongoing refinement through successes and setbacks. His adaptability—shifting airtimes and tailoring content—illustrates continuous growth through reflection. By blending radio and SAWBO animations, Juma cultivates lifelong learning in rural agriculture. Farmers actively engage with evolving knowledge, continuously adapting their practices to boost productivity and improve livelihoods. Lifelong learning—practiced by Juma and his audience—goes beyond acquiring knowledge to continuous reflection and adaptation. It strengthens skills for a shifting agricultural landscape and fosters dynamic knowledge exchange between change agents and communities. Juma's blend of radio and SAWBO animations shows how analog and visual tools fuel lifelong learning in rural Kenya. His adaptive, reflective approach empowers farmers and builds a learning feedback loop. In areas with limited digital access, this strategy bridges the divide and makes agricultural knowledge widely available.

17.4 Recommendations and Future Directions

Blending analog tools—like radio, printed materials, and farmer gatherings—with digital learning strategies is essential for bridging rural digital divides [34]. While digital ICTs hold promise, analog channels remain vital for reaching underserved populations, especially women farmers [21]. Community-based formats, including field schools, enable peer learning and hands-on adoption of new methods [35]. Integrating these with digital tools like SAWBO animations enhances agricultural education. Yet, financial and literacy barriers hinder digital uptake, underscoring the need for gender-sensitive policies and inclusive, community-driven approaches [10]. AI can help close digital gaps—especially with improved rural access to internet and smartphones. It also creates space for women to shape gender-sensitive content and build digital literacy. When used as a tool, AI holds promise for addressing the gender digital divide.AI remains bound by digital infrastructure, reinforcing barriers faced by women and rural populations. To prevent deepening digital divides, analog methods—like community gatherings—must complement AI use. Without inclusive access, AI risks further marginalizing vulnerable groups.

17.5 Conclusion

This chapter explores how blending analog and digital tools—especially mobile technologies—can help bridge the gender digital divide in rural Kenya. It examines barriers women face in accessing digital tools, highlights mobile strategies for promoting gender equity aligned with SDG 5, and emphasizes the complementary role of analog methods in advancing inclusive digital solutions.

This chapter highlights how gender-responsive digital policies in Kenya can boost economic growth, social mobility, and governance. Digital inclusion gives women access to agricultural, health, political info, and mobile financial services like M-Pesa. However, ongoing disparities risk widening inequality and limiting digital gains. Expanding digital tools also supports SDGs by promoting lifelong learning, capacity-building, and adult education.

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